

TABLE II
Truth Table of Toffoli Gate

Input 1	Input 2	Input 3	Output 1	Output 2	Output 3
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	1
1	1	1	1	1	0

TABLE III
Truth Table of Fredkin Gate

Input 1	Input 2	Input 3	Output 1	Output 2	Output 3
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	0
1	1	1	1	1	1

TABLE IV
Mapping of Controlled Swapped Gate

Input 1	Input 2	Input 3	Output 1	Output 2	Output 3
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	0	1
1	1	1	1	1	1

quick factorisation of large numbers (an explicit threat to conventional cryptographical techniques), more accurate simulation of quantum phenomena and very efficient database search. For any search space of size ' n ' nodes, where each node represents a possible solution to a problem, there is only one possible solution, and each node must be checked individually for properties that correspond to a correct solution. Quantum computing offers dramatic speedup. In conventional computers, the average search time

is the time taken to check each node times the number of nodes (n) divided by two (it's probable that the solution will be found about halfway through the search). In quantum computers, the average search time is the time taken to check each node times the square root of ' n '. This confers a huge advantage, which becomes more significant for larger problems.

The idea and concept of quantum computing was introduced way back in 1970s and 1980s, by Richard Feynmann, David Deutsch and Paul Benioff. The first breakthrough in the concept was Peter Shor's factoring algorithm reported in 1994, followed by Lov Grover's searching algorithm in 1996. Isaac L. Chuang developed the world's first 2-qubit, 3-qubit, 5-qubit and 7-qubit quantum computers in 1996-2000.

The main features of quantum computing revolve around:

1. Quantum nature: a combination of both the qubits
2. In preparing the initial state: only one of the two qubits
3. On measurement: only one state found
4. Probability: the state's component in the mix
5. Both preparation and measurement in contact with a macro system

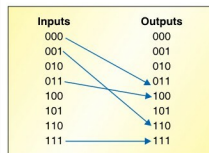
The basics of quantum computing start with preliminary knowledge of quantum beam splitting, quantum interference and quantum entanglement.

Quantum gate

A quantum gate is a quantum circuit that operates on a number of qubits. Characteristics of qubits as mentioned above make quantum gates different from logic gates in two respects: reversibility and quantum parallelism. The classical NOT gate is a reversible gate equivalent of quantum NOT gate.

For no power dissipation in the circuit, it is necessary that the arbitrary circuit be built from reversible gates. In reversible gates:

1. Feedback is not allowed
2. Fan-out is not allowed
3. Operation is a permutation of vectors as shown below:



Universal reversible quantum gates. The gates that could be used to make any logic circuit are called universal gates. In classical computers, NAND and NOR are universal gates. In quantum computers, three universal and reversible gates are Toffoli gate, Fredkin gate and controlled NOT gate.

Toffoli gate (controlled-controlled NOT gate (C-CNOT)). This is a 3-input, 3-output gate with truth table as shown in Table II. The first two outputs are copies of the first two inputs. The third output is XOR of the third input and AND of first two inputs. Toffoli gate may be seen as a modified controlled NOT gate.

Fig. 2 shows the functional diagram of Toffoli gate. A three-qubit Toffoli gate is known as Hadamard gate.

Fredkin gate. This is a three-input, three-output gate with truth table as shown in Table III. The first output is the copy of the first input. The last two outputs are swapped to match the last two inputs if the first input is 0. Other-